Lecture 13

De Rham cohomology

Objectives. *A closed form of positive degree has local primitives, forms whose exterior derivative is the given form, but a global primitive may not exist. De Rham cohomology records this obstruction. We prove the local assertion by integrating along a contraction. We then compare homotopic maps and compute the cohomology of spheres by joining two open sets. The Coulomb form gives a nonzero class with a measurable period.*

By the end of this lecture you will be able to:

- Distinguish closed forms, exact forms and their cohomology classes.
- Construct a primitive of a closed positive-degree form on a star-shaped open set.
- Prove smooth homotopy invariance using the homotopy operator.
- Compute sphere cohomology from Mayer–Vietoris and normalize its area class.

13.1 Closed forms modulo exact forms

All manifolds in this lecture are smooth and without boundary. All vector spaces and cohomology groups have real coefficients. Recall from Lecture 10 that a form α is *closed* if $\mathrm{d}\alpha = 0$ and *exact* if $\alpha = \mathrm{d}\eta$ for some form η. The identity $\mathrm{d}^2 = 0$ makes every exact form closed. A *cochain complex* is a sequence of vector spaces with linear maps $\mathrm{d}\colon C^k \to C^{k+1}$ whose successive composites are zero. The spaces $\Omega^k(M)$ with exterior differentiation form such a complex. We put $\Omega^k(M) = 0$ for $k < 0$.

The *de Rham cohomology* in degree k is the quotient

$$H^k(M) = \frac{\ker(\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M))}{\operatorname{im}(\mathrm{d}\colon \Omega^{k-1}(M) \to \Omega^k(M))}.$$

The *cohomology class* $[\alpha]$ of a closed form consists of all closed forms differing from α by an exact form. In positive degree, $[\alpha] = 0$ means precisely that α has a global primitive. In degree zero, only the zero function represents the zero class. The superscript distinguishes cohomology from Lecture 6's homology. The only exact form of degree zero is zero. A function is closed exactly when all its partial derivatives vanish. It is then constant on connected coordinate balls, hence locally constant. Conversely a locally constant function has zero derivative. In particular, $H^0(M) = \mathbb{R}$ when M is connected and nonempty.

For example, on the punctured plane the angular form is

$$\alpha_{\mathrm{ang}} = \frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2 + y^2}.$$

Lecture 10 proved that this form is closed and has integral 2π around the counter-clockwise unit circle. An exact form has zero integral around that circle by Stokes. Hence $[\alpha_{\mathrm{ang}}] \neq 0$.

More generally, a *period* of a closed k-form is its integral over a closed oriented k-manifold mapped smoothly into M, using pullback. Such a mapped manifold is a geometric cycle. Stokes makes every period of an exact form zero, so periods depend only on the class. These geometric cycles suffice here. Lecture 14 relates forms to the singular cycles of Lecture 6.

A smooth map $f\colon M \to N$ induces a linear map $f^*\colon H^k(N) \to H^k(M)$ by $f^*[\alpha] = [f^*\alpha]$. Indeed, pullback commutes with d, so it preserves closed forms and exact differences. Pullback reverses composition: $(g \circ f)^* = f^* g^*$, and the identity map induces the identity. The wedge product also descends to classes. For closed β, $(\mathrm{d}\eta) \wedge \beta = \mathrm{d}(\eta \wedge \beta)$. For closed α of degree k, $\alpha \wedge \mathrm{d}\xi = (-1)^k \mathrm{d}(\alpha \wedge \xi)$. Changing either representative therefore changes the product only by an exact form. We write $H^*(M)$ for all the degrees together.

13.2 The homotopy operator and the Poincaré lemma

A *smooth homotopy* between smooth maps F_0 and F_1 from M to N is a smooth map $F\colon [0,1] \times M \to N$ with $F_t(p) = F(t,p)$ and those endpoint maps. Smoothness at the endpoints means local smooth extension, as in Lecture 12. No inverse for any F_t is required. For a k-form α on N, the pulled-back form decomposes uniquely as

$$F^*\alpha = \mathrm{d}t \wedge a_t + b_t,$$

where a_t and b_t are forms on M of degrees $k-1$ and k. Here a_t is the restriction of $\iota_{\partial_t} F^*\alpha$ to the slice at time t, and $b_t = F_t^*\alpha$. Define the *homotopy operator* by integrating its coefficients:

$$K_F\alpha = \int_0^1 a_t \,\mathrm{d}t \quad (k \geq 1), \qquad K_F f = 0 \quad (k = 0).$$

The integral is smooth. Near each point, compactness of the time interval permits differentiating the smooth coefficients under the integral. The intrinsic definition by contraction makes the chart definitions agree.

Proposition 13.1 (Homotopy operator identity)**.** *For every smooth homotopy F and every differential form α,*

$$F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha + K_F\mathrm{d}\alpha.$$

Proof. Write d_M for differentiation in M. The product rule gives

$$\mathrm{d}(F^*\alpha) = \mathrm{d}t \wedge (\partial_t b_t - \mathrm{d}_M a_t) + \mathrm{d}_M b_t.$$

The minus sign comes from moving d_M past $\mathrm{d}t$. Since $\mathrm{d}F^*\alpha = F^*\mathrm{d}\alpha$, the time-containing coefficient on the right is the one integrated in $K_F\mathrm{d}\alpha$. Thus

$$K_F\mathrm{d}\alpha = \int_0^1 \partial_t b_t \,\mathrm{d}t - \mathrm{d}_M \int_0^1 a_t \,\mathrm{d}t = b_1 - b_0 - \mathrm{d}_M K_F\alpha.$$

The fundamental theorem of calculus gives the endpoint difference. This is the claimed identity. For functions, the same computation has $a_t = 0$ and is the ordinary endpoint formula. ∎

Two maps of complexes whose difference is $\mathrm{d}K + K\mathrm{d}$ are called *cochain homotopic*. The displayed identity constructs that relation for the two endpoint pullbacks.

An open set $U \subset \mathbb{R}^n$ is *star-shaped about* 0 if $0 \in U$ and $tx \in U$ for every $x \in U$ and $t \in [0,1]$. The figure shows the segments used by its contraction.

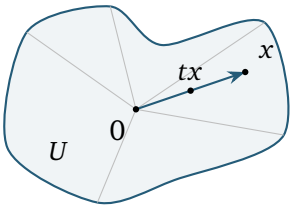


Figure 13.1: In a star-shaped set, each segment from 0 to x stays inside U. The contraction is given by $F(t,x) = tx$.

For this homotopy the operator has the explicit formula, for $k \geq 1$,

$$(K\alpha)_x(v_1,\dots,v_{k-1}) = \int_0^1 t^{k-1}\alpha_{tx}(x,v_1,\dots,v_{k-1})\,\mathrm{d}t. \tag{13.1}$$

Indeed, the time derivative of F is x, while each spatial tangent vector is multiplied by t. This gives the power t^{k-1}. Near a fixed $x \in U$, compactness of its segment and openness of U give $\varepsilon > 0$ and a neighbourhood of x such that $tx' \in U$ for every point x' in that neighbourhood and $t \in (-\varepsilon, 1+\varepsilon)$. This gives the required local smooth extension in time. Differentiating the coefficients under the integral therefore defines a smooth form on U.

Theorem 13.2 (Poincaré lemma). *On a star-shaped open set, every closed k-form with $k \geq 1$ is exact. Such a set has degree-zero cohomology consisting of the constant functions.*

Proof. After translating the star centre, use $F(t,x) = tx$. The map F_1 is the identity. The map F_0 is constant, so its differential is zero and its pullback on positive-degree forms is zero. For closed α the operator identity consequently gives $\alpha = \mathrm{d}K\alpha$. In degree zero it instead gives $f - f(0) = K\mathrm{d}f$. A closed function is therefore constant. ∎

For example, $\alpha = 2x\mathrm{d}x + 2y\mathrm{d}y$ on $\mathbb{R}^2$ has $K\alpha = \int_0^1 2t(x^2+y^2)\,\mathrm{d}t = x^2 + y^2$. The punctured plane is not star-shaped. Its radial paths end at a missing point, so the homotopy does not have its claimed target there. A formal insertion into α_{ang} would even give zero, because $-yx + xy = 0$. It cannot supply a primitive of that nonzero form.

13.3 Homotopy invariance

Theorem 13.3 (Homotopy invariance · *proved given cited smoothing; smooth homotopies proved directly*). *Smoothly homotopic maps induce the same maps on de Rham cohomology. The same holds for smooth maps that are continuously homotopic. Smooth manifolds with the same continuous homotopy type have isomorphic cohomology in every degree.*

Proof. For a smooth homotopy and a closed form α, the operator identity gives $F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha$. The two pullbacks therefore represent the same class.

For the continuous case we use the following smooth approximation results. Every continuous map between smooth manifolds is homotopic to a smooth map. Two homotopic smooth maps admit a smooth homotopy. These are cited from Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143. Choose a smooth representative of a continuous map's homotopy class and use its pullback. Two choices induce the same map by the second cited result and the smooth proof above. Composition and identities are preserved because compositions of smooth representatives represent the continuous compositions. If continuous maps $f\colon M \to N$ and $g\colon N \to M$ are homotopy inverses, their

composites represent identity maps. Contravariance then gives $f^*g^* = \mathrm{id}$ and $g^*f^* = \mathrm{id}$ on the respective cohomology spaces. Thus these maps are inverse isomorphisms. ∎

A contractible manifold consequently has the cohomology of a point: $\mathbb{R}$ in degree zero and zero in positive degrees. For $n \geq 1$, the punctured space $\mathbb{R}^n \setminus \{0\}$ retracts onto S^{n-1} by $r(x) = x/\|x\|$. With i the inclusion, $r \circ i = \mathrm{id}$. A smooth homotopy from the identity to $i \circ r$ is

$$F(t,x) = \left((1-t)^2 + \frac{t^2}{\|x\|}\right) x.$$

The coefficient is positive for every real t, since its two nonnegative summands never vanish together. Hence the homotopy stays in the punctured space. It has the required endpoints. This proves $H^*(\mathbb{R}^n \setminus \{0\}) \cong H^*(S^{n-1})$.

13.4 Exact sequences and Mayer–Vietoris

A sequence of vector spaces and linear maps is *exact* at a term if the image of the incoming map equals the kernel of the outgoing map. A *short exact sequence* has the form $0 \to A \to B \to C \to 0$. Its first nonzero map is injective and its last is surjective.

Let $M = U \cup V$ be an open cover and set $W = U \cap V$. Restriction gives a short exact sequence of form spaces in each degree:

$$0 \longrightarrow \Omega^k(M) \xrightarrow{i} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{j} \Omega^k(W) \longrightarrow 0, \qquad j(a,b) = b|_W - a|_W.$$

The map i restricts a global form to both sets. It is injective because these sets cover M. A pair is in $\ker j$ exactly when its forms agree on the overlap, which is precisely when they glue to a global form. To prove surjectivity, choose a partition $\rho_U + \rho_V = 1$ subordinate to the cover, supplied by Theorem 12.1. For $\zeta \in \Omega^k(W)$ take

$$a = -\operatorname{ext}_U(\rho_V \zeta), \qquad b = \operatorname{ext}_V(\rho_U \zeta).$$

Here ext means extension by zero from the overlap. This extension is smooth: near a point of $U \setminus V$, the support condition makes ρ_V zero, and similarly on $V \setminus U$. Their difference on W is $(\rho_U + \rho_V)\zeta = \zeta$. All restriction and difference maps commute with differentiation.

For a closed overlap form ζ, choose any lifts with $j(a,b) = \zeta$. Then $\mathrm{d}b - \mathrm{d}a = \mathrm{d}\zeta = 0$ on W. The derivatives glue to a global closed form c of degree $k+1$. The *connecting homomorphism* is

$$\delta_{\mathrm{MV}} \colon H^k(W) \longrightarrow H^{k+1}(M), \qquad \delta_{\mathrm{MV}}[\zeta] = [c], \quad c|_U = \mathrm{d}a, \quad c|_V = \mathrm{d}b.$$

There is no additional sign. The subscript identifies this as the Mayer–Vietoris map. Different lifts differ by restrictions of a global form η, so their resulting c differs by $\mathrm{d}\eta$. If ζ changes by $\mathrm{d}\xi$, lift ξ to (a', b') and replace (a,b) by $(a + \mathrm{d}a', b + \mathrm{d}b')$. The resulting c is unchanged. In degree zero there is no such exact change. These observations prove well-definedness. Adding or scaling lifts proves linearity.

Theorem 13.4 (Mayer–Vietoris · *exactness sketched; connecting map and overlap exactness proved given partitions of unity*)**.** *The induced maps and connecting homomorphisms form an exact sequence*

$$\begin{aligned} 0 \longrightarrow H^0(M) \xrightarrow{i^*} H^0(U) \oplus H^0(V) \xrightarrow{j^*} H^0(W) \xrightarrow{\delta_{\mathrm{MV}}} H^1(M) \longrightarrow \cdots, \\ \cdots \longrightarrow H^k(M) \xrightarrow{i^*} H^k(U) \oplus H^k(V) \xrightarrow{j^*} H^k(W) \xrightarrow{\delta_{\mathrm{MV}}} H^{k+1}(M) \longrightarrow \cdots. \end{aligned} \tag{13.2}$$

Proof at the overlap term and sketch elsewhere. If $\zeta = b - a$ for closed lifts, its connecting form is zero. Hence $\operatorname{im} j^* \subset \ker \delta_{\mathrm{MV}}$. Conversely, if $\delta_{\mathrm{MV}}[\zeta] = 0$, its glued form is $c = \mathrm{d}\eta$ for a global η. Subtract the restrictions of η from both lifts. Their derivatives now vanish and their difference remains ζ. Thus $[\zeta] \in \operatorname{im} j^*$, proving exactness at $H^k(W)$.

At the pair term, an exact difference of closed representatives can be removed by lifting its primitive through j. The adjusted representatives agree and glue. At the global term, local primitives of a class restricting to zero have a closed difference on W whose connecting class is the original class. These are the other exactness arguments; their representative and degree-zero details are assigned to Sheet 13. This is the cochain construction of Tu, §25–§26, especially (26.1), pp. 288–291. ∎

13.5 Cohomology of spheres

Remove the south pole to obtain $U \subset S^n$ and the north pole to obtain V, for $n \geq 1$. Each set is diffeomorphic to $\mathbb{R}^n$ by stereographic projection. Their overlap has the product description in the figure.

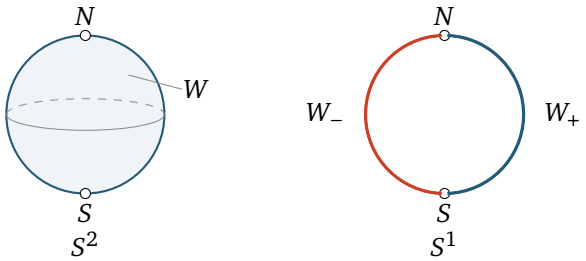


Figure 13.2: Each open set omits one pole. Their overlap is the whole sphere except the two marked poles, an open cylinder for S^2. For S^1 it has two components.

Write a sphere point as $(x', z) \in \mathbb{R}^n \times \mathbb{R}$. On W, the map $(x', z) \mapsto (x'/\sqrt{1 - z^2}, z)$ is a diffeomorphism onto $S^{n-1} \times (-1, 1)$, with inverse $(u, z) \mapsto (\sqrt{1 - z^2}u, z)$. Contracting the second coordinate to zero retracts W onto S^{n-1}. The two contractible cover members have zero positive-degree cohomology.

Theorem 13.5 (Sphere computation · *proved given Mayer–Vietoris exactness*)**.** *For $n \geq 1$ and $k \geq 0$,*

$$H^k(S^n) \cong \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n, \\ 0, & \text{otherwise.} \end{cases}$$

For S^0, degree zero is $\mathbb{R}^2$ and all higher degrees vanish. For $n \geq 1$, punctured $\mathbb{R}^n$ has the cohomology of S^{n-1}.

Proof. The space S^0 is two points. A function assigns one value to each point, and all positive-degree forms vanish. This proves the initial case. For $n = 1$, the overlap has two contractible components. The beginning of Mayer–Vietoris is

$$0 \longrightarrow H^0(S^1) \longrightarrow \mathbb{R}^2 \xrightarrow{(a,b)\mapsto(b-a,b-a)} \mathbb{R}^2 \longrightarrow H^1(S^1) \longrightarrow 0.$$

The middle map has a one-dimensional kernel and image. Its cokernel therefore also has dimension one. Exactness gives both claimed groups. Higher degrees vanish because the circle has dimension one.

For $n \geq 2$, the overlap is connected, so its degree-zero group is $\mathbb{R}$. The difference map $\mathbb{R}^2 \to \mathbb{R}$ is surjective with diagonal kernel. Exactness gives $H^0(S^n) = \mathbb{R}$ and $H^1(S^n) = 0$. For $k \geq 2$, the zero positive-degree groups of U and V give

$$0 \longrightarrow H^{k-1}(W) \xrightarrow{\delta_{\mathrm{MV}}} H^k(S^n) \longrightarrow 0.$$

Thus $H^k(S^n) \cong H^{k-1}(S^{n-1})$. Induction gives precisely one nonzero positive-degree group, in degree n. The explicit retraction in § 13.3 proves the punctured-space assertion. In particular, punctured $\mathbb{R}$ has two-dimensional degree-zero cohomology, whereas punctured $\mathbb{R}^n$ for $n \geq 2$ is connected. ∎

13.6 The area class and local primitives on the sphere

On the outward-oriented unit sphere let β be the area form, the restriction of Lecture 12's Coulomb flux form. We compute its class using the ordered cover above: $U = S^2 \setminus \{\text{south}\}$ and $V = S^2 \setminus \{\text{north}\}$. Write the local 1-forms as a_U and a_V, and their common derivative as c. Their difference and derivatives are related as in the diagram.

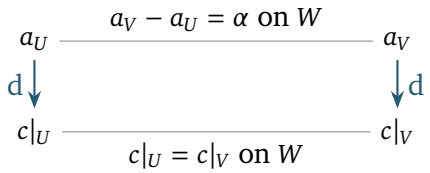


Figure 13.3: Local 1-forms differ by a closed form on W. Their derivatives agree and glue to the 2-form representing the connecting class.

Example 13.6 (The normalized sphere class)**.** The sphere computation gives $H^2(S^2) = \mathbb{R}$ and $H^1(S^2) = 0$. The form $\beta/(4\pi)$ is closed, since every 3-form on a surface vanishes. Its period is

$$\int_{S^2} \frac{\beta}{4\pi} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = 1.$$

It is therefore not exact and its class spans $H^2(S^2)$.

We can also compute the sign of the connecting map. On W, the globally defined angular 1-form is $\alpha = \mathrm{d}\phi/(2\pi)$, where $\mathrm{d}\phi$ denotes the angular form, not the differential of a globally defined real angle. Choose lifts

$$a_U = -\frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1+z)}, \qquad a_V = \frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1-z)}.$$

Their denominators do not vanish on their respective domains, so these forms are smooth also at the opposite poles. On the overlap, $x\mathrm{d}y - y\mathrm{d}x = (1-z^2)\mathrm{d}\phi$. Consequently $a_U = -(1-z)\mathrm{d}\phi/(4\pi)$ and $a_V = (1+z)\mathrm{d}\phi/(4\pi)$, giving $a_V - a_U = \alpha$. Their common derivative is

$$\mathrm{d}a_U = \mathrm{d}a_V = \frac{\mathrm{d}z \wedge \mathrm{d}\phi}{4\pi} = -\frac{\sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\phi}{4\pi} = -\frac{\beta}{4\pi}.$$

Equality extends over each pole by smoothness. Thus $\delta_{\mathrm{MV}}[\mathrm{d}\phi/(2\pi)] = -[\beta/(4\pi)]$, with the frozen order of U and V. The positive period-one class is $[\beta/(4\pi)]$. A nonzero period obstructs joining its local primitives to a global one. Write β_C for the closed Coulomb form of Lecture 12 on $\mathbb{R}^3 \setminus \{0\}$. Its restriction to S^2 is the area form β. Restriction is an isomorphism on cohomology, inverse to the radial retraction pullback. It sends $[\beta_C/(4\pi)]$ to $[\beta/(4\pi)]$. Hence $[\beta_C/(4\pi)]$ spans $H^2(\mathbb{R}^3 \setminus \{0\})$, with sphere period one.

Real coefficients lose torsion information. A torsion element is one annihilated by a positive integer. No real vector space has a nonzero torsion element. In particular, Lecture 4's order-two loop in $\mathrm{SO}(3)$ cannot be detected by a closed real 1-form's period. Use smooth loops constant near their endpoints, so their concatenation is smooth. Periods of a closed 1-form are invariant under smooth loop homotopy by Stokes on the cylinder. Continuous homotopies can be smoothed by the cited theorem. Periods add under loop composition, so twice that loop's period is zero and its real period is zero. This does not impose any integer normalization on the sphere class. Lecture 22 obtains that extra information from a bundle.

Self-check. Find $H^1(\mathbb{R}^2 \setminus \{0\})$ and a generator of period one on the counter-clockwise unit circle.

Answer. Radial retraction gives the cohomology of S^1, so this space is one-dimensional. The class $[\alpha_{\mathrm{ang}}/(2\pi)]$ has period one by Lecture 10. It is nonzero and therefore spans the space.

Summary. De Rham cohomology consists of closed forms modulo exact forms, over $\mathbb{R}$. The homotopy operator proves local exactness in positive degree on star-shaped sets and invariance under smooth homotopy. Cited smoothing extends invariance to continuous homotopy type. Mayer–Vietoris computes sphere cohomology from two contractible sets. On S^2, the area class $[\beta/(4\pi)]$ has period one. Lecture 14 relates periods to degree and singular homology.

Reading. Tu, *An Introduction to Manifolds*, §24–§29, especially §26, pp. 288–291, and §29, pp. 311–317. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143.